

Jesús M. Rueda-Becerril, Ph.D.

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Professional Summary

Computational Scientist with a PhD in Physics and 10+ years developing high-performance numerical algorithms and physics-based simulations. Expertise in scientific machine learning and geospatial data analysis. Strong mathematical foundations with a track record of architecting open-source simulation libraries, leading funded research projects, and translating complex physical models into production-quality scientific software. Experienced working across cross-functional research and engineering teams.

Skills

Languages: Python, C/C++, Fortran 95, R, SQL, Shell, Rust, Java, Julia

ML & Data: Scientific ML, PyTorch, TensorFlow, scikit-learn, NumPy, SciPy, Pandas, Astropy, ML-assisted parameter estimation, gradient-based optimization, SVM, random forest, neural networks

Geospatial & Remote Sensing: Sentinel-1/2 multispectral processing, raster/vector data pipelines, QGIS, geospatial feature engineering for ML

HPC & Parallel: OpenMP, MPI, OpenACC, SLURM, PBS, HDF5, large-scale parallel CPU computing

Simulation: Physics simulation (multiscale/multiphysics modeling, radiative transfer, relativistic MHD, Lagrangian particle dynamics), numerical PDEs (advection-diffusion, Fokker-Planck), finite difference/volume methods, numerical linear algebra, synthetic dataset generation, uncertainty quantification, parameter estimation

Tools: Git, Docker, Azure, Jenkins, Kafka, Jira, QGIS, ParaView, VisIt

Experience

Independent Contractor

Jan 2026 – Present

Mercor

Remote

- Physics expert for a leading AI lab: reviewed 50+ expert-level problems in physics, hydrodynamics, astrophysics, and cosmology; wrote original prompts that stumped frontier models.
- Diagnose failure modes in model reasoning and write structured feedback to improve scientific accuracy.

Spatial Data Scientist

May 2025 – Jan 2026

TealWaters

Seattle, WA

Built geospatial ML pipelines and optimized terrain-analysis code for wetland mapping.

- Mapped wetland probability across the 2,160 km² Skykomish watershed with the Wetland Intrinsic Potential (WIP) random-forest model, reaching >75% accuracy.
- Optimized the terrain-modeling code that computes elevation derivatives from 1 m, 3 m, and 10 m DEMs over 20 tiles, improving its computational performance.
- Engineered features from Sentinel-1/2 multispectral imagery and terrain derivatives for ML land-cover and wetland models; built Python tools and QGIS visualizations.
- Presented results to managers and decision-makers, trained 2 colleagues on the WIP tool, and worked with software engineers to move prototype tools into production.

Independent Researcher & Open-Source Developer

Jan 2024 – May 2025

Self-directed

Seattle, WA

Continued simulation research and open-source development between industry roles.

- Co-developed Tleco, a Rust/Python toolkit that models radiation from relativistic outflows by solving the Fokker-Planck equation coupled to radiative transfer; published in ApJ (2024) [9].
- Developing WindsOfChange with the UW Forest Mycobiome Lab: a C++/R library that simulates wind-driven spore dispersal over forested, hilly terrain, parallelized with RcppParallel and OpenMP.
- Invited talk on machine learning in environmental science at the University of Washington (2024).

Software Engineer

Apr 2022 – Jan 2024

Paychex

Remote

Built and tested data-integration services between enterprise systems and Oracle E-Business Suite.

- Built Java Kafka consumers that moved data between enterprise databases, deployed through Jenkins and OpenShift CI/CD pipelines.
- Created Splunk dashboards tracking the 200–400 users shared between enterprise systems and Oracle EBS, with alerts for production incidents; prepared and validated Oracle EBS data with SQL.
- Wrote full test suites for 5+ Python services in the CI pipeline, keeping production defect rates low.

Postdoctoral Research Associate*Rochester Institute of Technology*

Feb 2021 – Apr 2022

Rochester, NY

HPC simulations of supermassive black hole binaries in an NSF-funded project.

- Led a 7-person team upgrading PatchworkMHD, a C code for magnetohydrodynamic simulations of supermassive black hole binaries, in an NSF-funded project.
- Added black hole spin to the simulations with under 5% runtime overhead.
- Ran and benchmarked large production simulations on Frontera (TACC), about 10,000 node-hours, and identified performance bottlenecks in the parallel code.
- Calibrated Paramo model parameters against Fermi-LAT gamma-ray spectra with gradient-based optimization, and supervised a graduate student applying gradient descent to the same problem.
- Co-authored 2 papers from a multi-institution study of binary neutron star mergers [6, 7] and a third on blazar heating and cooling [8].

Postdoctoral Research Fellow*Purdue University*

Oct 2018 – Nov 2020

West Lafayette, IN

Radiative-transfer modeling of relativistic jets (blazars and gamma-ray burst afterglows).

- Architected Paramo, an open-source Fortran 95 library that solves the Fokker–Planck equation coupled to radiative transfer; parallelized and optimized it with OpenMP, making it about 20× faster than the original serial code on 5 cores.
- Paramo has been adopted by other research groups, including Combi & Siegel (2023).
- Co-I on two NASA Fermi Guest Investigator proposals; one was funded (\$68K) and supported the finding that blazar jets are launched with similar energy per baryon, independently of their power [5].
- Added numerical-integration and OpenMP features to Paramo and built Python tools to compute energy losses and spectra of high-energy particles in gamma-ray burst afterglows [4].
- Mentored 3 graduate students (1 M.S., 2 Ph.D.), co-authoring papers with each.
- Modeled the COVID-19 outbreak in Mexico (Python) and co-wrote Spanish-language infographics in a campaign against misinformation.

Postdoctoral Research Fellow*Universidad Michoacana de San Nicolás de Hidalgo*

Jan 2018 – Sep 2018

Morelia, Mexico

- Built a Python pipeline that turned GRTrans black hole simulation images into training data for an SVM classifier of black hole spin (grayscale conversion, intensity normalization, spatial feature extraction).
- Released an open-source Python tool for radiative-transfer spectra and light curves.
- Organized an HDF5 workshop to train graduate students in high-volume data storage.

Graduate Research Assistant*Universitat de València*

Oct 2011 – Jul 2017

Valencia, Spain

- Automated runs of the C-SPEV radiative-transfer code with Shell and Python pipelines: 10–60 simulations per project and tens of GB of HDF5 output, with data-quality checks for downstream analysis.
- Analyzed Fermi-LAT and VLBA multi-wavelength observations, cross-matching sources across catalogs and fitting log-parabola spectra.
- Fit C-SPEV simulation models to telescope data, identifying spectra and light-curve patterns that quantify plasma magnetization in blazars.
- Extended C-SPEV to compute discrete and continuous spectra from particle distributions of arbitrary shape, without affecting simulation runtime.
- Published 2 first-author papers in MNRAS [2, 3].

Education**Ph.D. in Physics***Universitat de València*

Oct 2011 – Jul 2017

Valencia, Spain

Excellent *cum laude*

Advisors: Prof. Miguel A. Aloy and Dr. Petar Mimica

Thesis: Numerical treatment of radiation processes in the internal shocks of magnetized relativistic outflows

M.Sc. in Physics*Universidad Michoacana de San Nicolás de Hidalgo*

Aug 2009 – Sep 2011

Morelia, Mexico

Advisor: Prof. José A. Cervera

Thesis: *Study of TOV stars with the SPH method*

B.Sc. in Physics

Universidad Autónoma del Estado de México

Advisor: Prof. Francisco S. Guzmán

Thesis: *Numerical solution of null geodesics for the generation of gravitational lenses produced by spherically-symmetric and static spacetimes*

Lic. Juan Josafat Pichardo Cruz Award

Aug 2004 – Dec 2008

Toluca, Mexico

Publications

Articles (9 published, 3 first-author; 3 in preparation)

- [12] Winkowski, S., **Rueda-Becerril, J. M.**, & Willing, C. *Using Numerical Simulations of Wind on Hilly Terrain to Model Propagule Dispersal*, in prep.
- [11] Holzman, L., **Rueda-Becerril, J. M.**, Creamer, C., & Willing, C. *Random Forest Link Landscape Position to Drought Resilience in Giant Sequoia Forests*, in prep.
- [10] **Rueda-Becerril, J. M.**, Winkowski, S., & Willing, C. *WindsOfChange: A New Toolkit for Modeling Spore Dispersal in Hilly Terrains*, in prep.
- [9] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Tleco: A Toolkit for Modeling Radiative Signatures from Relativistic Outflows*, ApJ 976, 182 (2024), arXiv:2405.17581.
- [8] Davis, Z., **Rueda-Becerril, J. M.**, & Giannios, D. *Balancing Turbulent Heating with Radiative Cooling in Blazars*, MNRAS 513, 5766–5779 (2022), arXiv:2201.07790.
- [7] Lopez-Armengol, F. G., Etienne, Z. B., [...], **Rueda-Becerril, J. M.**, [...] *Handing off the outcome of binary neutron star mergers for accurate and long-term postmerger simulations*, Phys. Rev. D 106, 083015 (2022), arXiv:2112.09817.
- [6] Murguía-Berthier, A., Noble, S., [...], **Rueda-Becerril, J. M.**, [...] *HARM3D+NUC: A New Method for Simulating the Post-merger Phase of Binary Neutron Star Mergers with GRMHD, Tabulated EOS, and Neutrino Leakage*, ApJ 919, 95 (2021), arXiv:2106.05356.
- [5] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *Blazar jets launched with similar energy per baryon, independently of their power*, MNRAS 501, 4092–4102 (2021), arXiv:2009.02273.
- [4] Zhang, H., Christie, I., Petropoulou, M., **Rueda-Becerril, J. M.**, & Giannios, D. *Inverse Compton Signatures of Gamma-Ray Burst Afterglows*, MNRAS 496, 974–986 (2020), arXiv:1910.14049.
- [3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *On the influence of a hybrid thermal–non-thermal distribution in the internal shocks model for blazars*, MNRAS 468, 1169–1182 (2017), arXiv:1612.06383.
- [2] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *The influence of the magnetic field on the spectral properties of blazars*, MNRAS 438, 1856–1869 (2014), arXiv:1310.5441.
- [1] Guzmán, F. S., & **Rueda-Becerril, J. M.** *Spherical boson stars as black hole mimickers*, Phys. Rev. D 80, 084023 (2009), arXiv:1009.1250.

Conference Proceedings

- [P5] **Rueda-Becerril, J. M.** *A numerical approach for radiative cooling in relativistic outflows*, Astron. Nachr. 342, 277–282 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13797.
- [P4] **Rueda-Becerril, J. M.**, Harrison, A. O., & Giannios, D. *The blazar sequence revised*, Astron. Nachr. 342, 147–152 (2021), 9th International Workshop on Astronomy and Relativistic Astrophysics, arXiv:2011.13805.
- [P3] **Rueda-Becerril, J. M.**, Mimica, P., & Aloy, M. A. *Numerical simulations of the internal shock model in magnetized relativistic jets of blazars*, PoS(SWIFT 10) 233, 159 (2014), arXiv:1502.07882.
- [P2] **Rueda-Becerril, J. M.**, Mimica, P., Aloy, M. A., & Aloy, C. *Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets of blazars*, EPJ Web Conf. 61, 02007 (2013), arXiv:1309.4612.
- [P1] Mimica, P., Aloy, M. A., **Rueda-Becerril, J. M.**, Tabik, S., & Aloy, C. *Numerical simulations of dynamics and emission from relativistic astrophysical jets*, J. Phys.: Conf. Ser. 454, 012001 (2013), arXiv:1211.1794.

Grants and Fellowships

Research Grants

NASA Fermi Guest Investigator Program, Cycle 12

2019

A simple model to understand the blazar sequence

Grant #121077, \$68K

PI: D. Giannios; Co-I: J. M. Rueda-Becerril

Fellowships

International Postdoctoral Fellowship

Oct 2018 – Nov 2020

National Council of Science and Technology (CONACyT), Mexico

Postdoctoral Fellowship

Jan 2018 – Sep 2018

Secretariat of Public Education, Mexico (Program for the Professional Development of Higher Education Institutions)

Study-Abroad Fellowship <i>National Council of Science and Technology (CONACyT), Mexico</i>	Sep 2014 – Aug 2016
Santiago Grisolia Fellowship <i>Council of Education, Research, Culture and Sport, Valencian Community, Spain</i>	Oct 2011 – Jun 2014
M.Sc. Fellowship <i>CONACyT, at the Institute of Physics and Mathematics, Universidad Michoacana de San Nicolás de Hidalgo</i>	Sep 2009 – Aug 2011
Summer Research Fellowship (XVII Summer of Scientific Research) <i>Mexican Academy of Sciences</i>	Jun 2007 – Aug 2007

Software and Projects

Blazar Population ML <i>Scientific ML (Python, scikit-learn, PyTorch)</i>	Jan 2026 – Present github.com/altjerue/ilwikak-ml
- Applying classical ML and Bayesian inference to the Fermi-LAT 4LAC-DR3 catalog (3,407 sources) to recover physical jet parameters from multi-wavelength data. - Building a neural network that maps physical jet parameters to observable spectral energy distributions accurately enough to compare with Tleco results. - Forecasting blazar gamma-ray flares with a time-series foundation model fine-tuned on Tleco-generated synthetic light curves.	
WindsOfChange <i>Lagrangian particle-dispersion library (C++/R)</i>	Aug 2024 – Present github.com/UW-ForestMycobiomeLab/WindSporesHills
Simulates particle dispersion, atmospheric turbulence, and flow over hilly terrain with forest canopy, based on analytical models (Massman & Weil 1999; Finnigan & Belcher 2004); designed to scale to tens of thousands of particles over kilometer-scale domains.	
Tleco <i>Radiative-transfer toolkit (Rust/Python)</i>	Jan 2024 – Sep 2024 github.com/zkdavis/Tleco
Numerical PDE solvers, particle-distribution algorithms, and spectral calculations for relativistic plasmas; successor to Paramo [9].	
Paramo <i>Radiative-transfer library (Fortran 95, OpenMP)</i>	Oct 2018 – Apr 2022 github.com/altjerue/paramo
Fokker–Planck and radiative-transfer solver for relativistic particle distributions; about 20× faster than the original serial code on 5 cores after OpenMP parallelization and optimization; used by other groups, including Combi & Siegel (2023).	
PatchworkMHD <i>Large-scale MHD simulation code (NSF project)</i>	Feb 2021 – Apr 2022
Led a 7-person upgrade of this production C code for HPC environments, adding black hole spin with under 5% runtime overhead.	

Awards

Marcos Moshinsky Award for Best Poster <i>9th International Workshop on Astronomy and Relativistic Astrophysics (IWARA 2020), video conference</i>	Sep 2020
Lic. Juan Josafat Pichardo Cruz Award <i>Universidad Autónoma del Estado de México</i> For defending a licentiate thesis within a year of completing the undergraduate coursework.	2009

Presentations

Invited Talks

The Role of Machine Learning in Environmental Science <i>Seminar on Climate Change Ecology, University of Washington</i>	Mar 2024 Seattle, WA
Morphology of the spectra from numerical simulations of the internal shocks model for blazars <i>Astrophysics Seminar, Purdue University</i>	Feb 2019 West Lafayette, IN
Numerical simulations of the internal shocks model in magnetized relativistic jets of blazars <i>DATA Group Weekly Seminar, Instituto de Astronomía, UNAM</i>	Jun 2018 Mexico City, Mexico
Numerical treatment of non-thermal radiation in the internal shocks model for blazars <i>Weekly Seminar, Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo</i>	Mar 2018 Morelia, Mexico

Numerical simulations of the internal shock model in magnetized relativistic jets of blazars <i>IVICFA's Fridays: Computation in Physics, IFIC</i>	Oct 2014 Paterna, Spain
Contributed Talks	
Simulations of supermassive binary black holes accretion dynamics in the spinning case <i>APS April Meeting 2022</i>	Apr 2022 New York, NY
A numerical approach to the Klein-Nishina corrections of radiative cooling in relativistic outflows <i>APS April Meeting 2021</i>	Apr 2021 Online
The blazar sequence revised <i>9th International Workshop on Astronomy and Relativistic Astrophysics</i> Video: youtube.com/watch?v=BAZNWLNT69M	Sep 2020 Online
Influence of the magnetic field on the spectral properties of blazars in the internal shocks scenario <i>Extreme-Astrophysics in an Ever-Changing Universe</i>	Jun 2014 Ierápetra, Greece
Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets <i>XXXIV Biennial Meeting of the Royal Spanish Society of Physics</i>	Jul 2013 Valencia, Spain
Posters	
A numerical approach for radiative cooling in relativistic outflows <i>9th International Workshop on Astronomy and Relativistic Astrophysics</i> Marcos Moshinsky Award for Best Poster. Video: youtube.com/watch?v=0TJiKg7kOPI	Sep 2020 Online
Numerical simulations of the internal shock model in magnetized relativistic jets of blazars <i>Swift: 10 Years of Discovery</i>	Dec 2014 Rome, Italy
Numerical study of broadband spectra caused by internal shocks in magnetized relativistic jets <i>The Innermost Regions of Relativistic Jets and Their Magnetic Fields</i>	Jun 2013 Granada, Spain
Teaching and Mentoring	
Zachary Davis, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored papers [8, 9]	2018 – 2022 Mentoring
Amanda O. Harrison, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored paper [5]	2018 – 2020 Mentoring
Hao Zhang, graduate student <i>Department of Physics and Astronomy, Purdue University</i> Co-authored paper [4]	2018 – 2019 Mentoring
Thermodynamics (graduate level) <i>Instituto de Física y Matemáticas, Universidad Michoacana de San Nicolás de Hidalgo</i>	Jun 2018 Substitute Instructor
Professional Development	
High Performance Computing on Frontera <i>Jason Allison et al., TACC, Austin, TX</i>	May – Jun 2021 Lecture series
Writing Winning Grants <i>Dr. Lauren Broyles, Purdue University</i>	Nov 2019 Lecture
XSEDE HPC Workshop: Summer Boot Camp <i>John Urbanic, Purdue University</i>	Jun 2019 Workshop
Data Analysis and Machine Learning with Python <i>Dr. Alejandro Torres, Universitat de València</i>	Feb 2017 Workshop
Numerical Relativity Simulations of BBH Coalescence Using the Einstein Toolkit <i>Dr. Vassilios Mewes, Universitat de València</i>	Jul 2016 Workshop, 8 hours
The Universe in the Light of PLANCK and BICEP2 <i>Prof. Nick Mavromatos, Universitat de València</i>	May 2014 Lecture series, 2 credits
Dark Matter <i>Prof. Alejandro Ibarra, Universitat de València</i>	Sep 2013 Lecture series, 2 credits

International Cargèse School on Cosmic Accelerators <i>Institut d'Études Scientifiques de Cargèse, France</i>	Apr – May 2013 Summer school
Introduction to C++ Programming <i>Dr. Jacek Generowicz, Universitat de València</i>	Apr 2012 Workshop, 6 credits
Numerical Relativistic Astrophysics <i>Prof. Luciano Rezzolla, Universitat de València</i>	Mar – Apr 2012 Lecture series, 9 hours
Fortran for Scientific Computing <i>HLRS, University of Stuttgart, Germany</i>	Mar 2012 Workshop, 33 hours

Outreach

Los más rápidos y los más furiosos (The Fastest and the Most Furious) <i>Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco</i>	Sep 2020 Online talk
Una simulación de la física y la astrofísica (A Simulation of Physics and Astrophysics) <i>Community of Undergraduate Physics Students, Juárez Autonomous University of Tabasco</i>	Aug 2020 Online talk
ANITA y la teoría de los universos paralelos (ANITA and the Theory of Parallel Universes) <i>Científicos Mexicanos en el Extranjero</i> mexiciencia.github.io/post/anita	May 2020 Blog post
¿Qué es el modelo SIR? (What is the SIR Model?) <i>Científicos Mexicanos en el Extranjero</i> mexiciencia.github.io/post/modelo-sir	May 2020 Blog post
Evolución del brote epidémico de COVID-19 (Evolution of the COVID-19 Epidemic Outbreak) <i>Científicos Mexicanos en el Extranjero</i> Data analysis and modeling collaborator. mexiciencia.github.io/post/covid19	Apr 2020 Blog post
Annual Department of Physics and Astronomy Poster Event <i>Purdue University</i>	Nov 2019 3 posters
Post-Doc Panel Q&A: What Happens When We Complete Our PhDs? <i>Department of Physics and Astronomy, Purdue University</i>	Apr 2019 Panelist
Annual Department of Physics and Astronomy Poster Event <i>Purdue University</i>	Nov 2018 Poster
¿Decía Einstein la verdad? (Was Einstein Telling the Truth?) <i>Facultad de Ciencias, Universidad Autónoma del Estado de México</i>	Mar 2009 Talk

Service

Organizing contributor <i>X Scientific Meeting of the Spanish Astronomical Society</i>	Dec 2012 Valencia, Spain
Student representative, Governing Council of the Faculty of Sciences <i>Universidad Autónoma del Estado de México</i>	Aug 2007 – May 2009 Toluca, Mexico

Languages

Spanish (native), English (full professional proficiency), Catalan (basic), Nahuatl (basic)